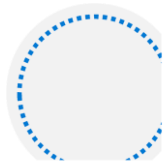


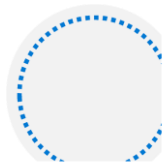
Training Course

Module 02: Governance and Compliance

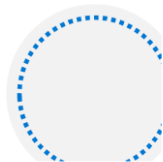




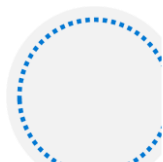
Lesson 01: Configure Subscriptions



Lesson 02: Configure Azure Policy



Lesson 03: Configure Role-Based Access Control



Lesson 04: Module 02 Lab

Lesson 01: Configure Subscriptions

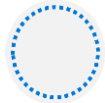




Identify Regions



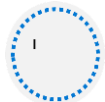
Implement Azure Subscriptions



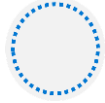
Obtain a Subscription



Identify Subscription Usage



Implement Cost Management



Apply Resource Tagging



Apply Cost Savings

Identify Regions

A region represents a collection of datacenters

Provides flexibility and scale

Preserves data residency

Select regions close to your users

Be aware of region deployment availability

There are global services that are region independent

Regions are paired for high availability



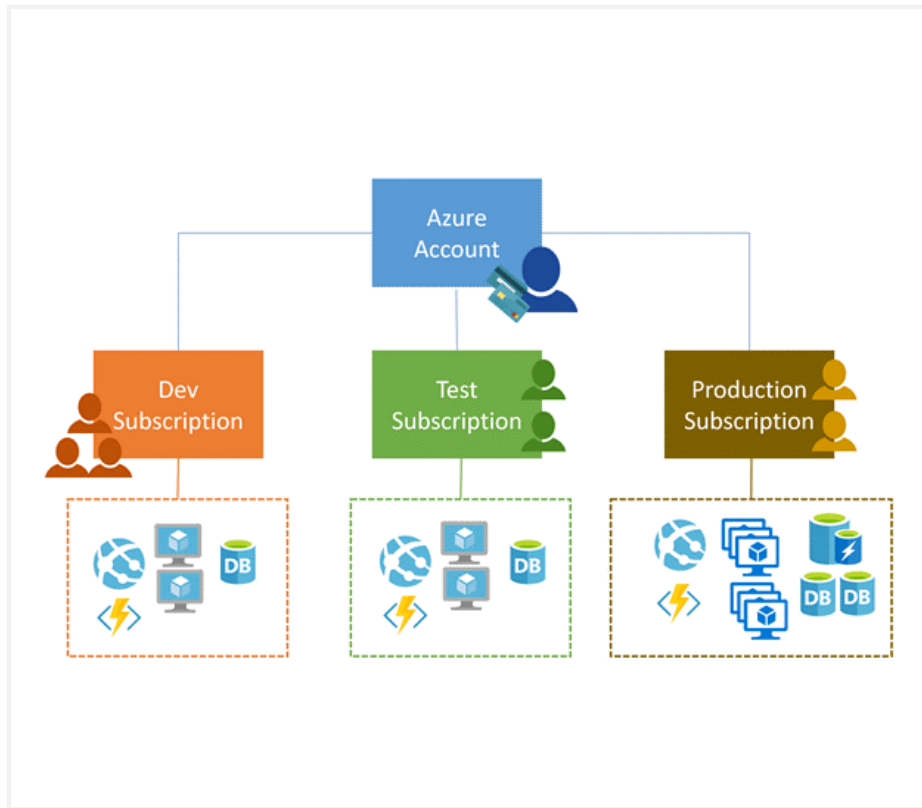
Worldwide there are 60+ regions
representing 140 countries

Implement Azure Subscriptions

Only identities in Azure AD, or in a directory that is trusted by Azure AD, can create a subscription

Logical unit of Azure services that is linked to an Azure account

Security and billing boundary



Obtain a Subscription

Enterprise Agreement customers make an upfront monetary commitment and consume services throughout the year

Resellers provide a simple, flexible way to purchase cloud services

Partners can design and implement your Azure cloud solution

Personal free account – Start right away



Identify Subscription Usage

Subscription	Usage
Free	Includes a \$200 credit for the first 30 days, free limited access for 12 months
Pay-As-You-Go	Charges you monthly
CSP	Agreement with possible discounts through a Microsoft Cloud Solutions Provider Partner – typically for small to medium businesses
Enterprise	One agreement, with discounts for new licenses and Software Assurance – targeted at enterprise-scale organizations
Student	Includes \$100 for 12 months – must verify student access

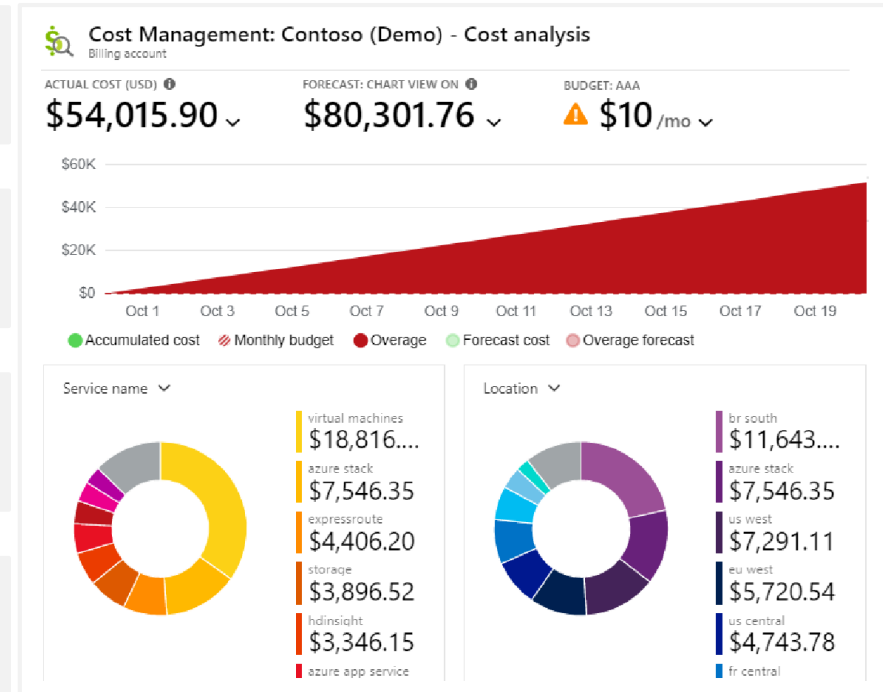
Implement Cost Management

Conduct cost analysis

Create a budget

Review recommendations

Export the data



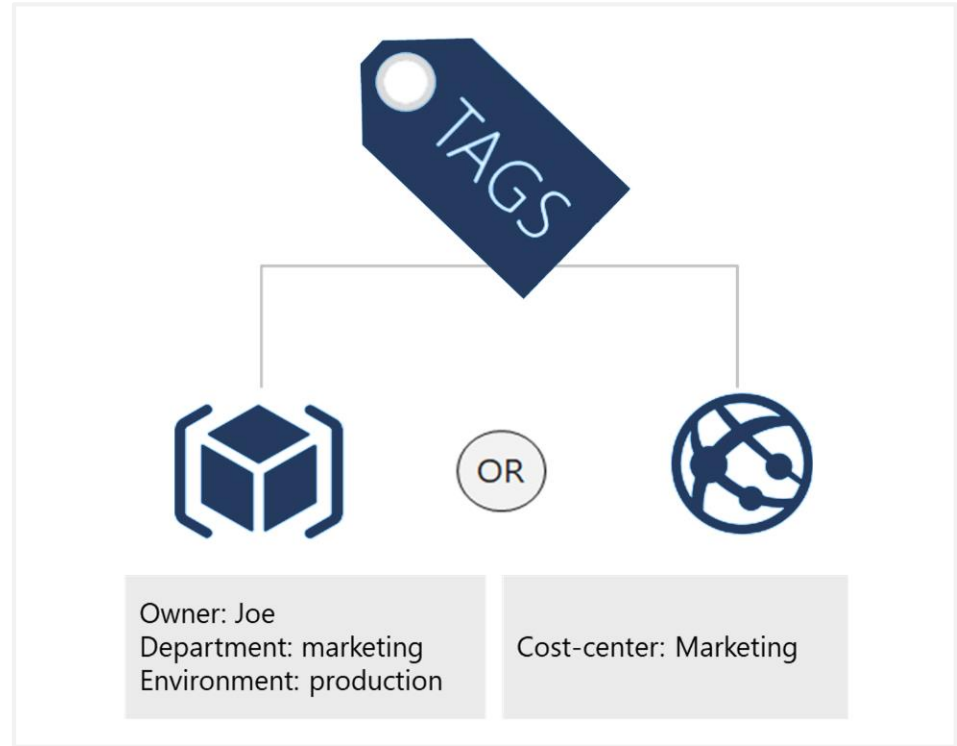
Apply Resource Tagging

Provides metadata for your Azure resources

Logically organizes resources into a taxonomy

Consists of a name-value pair

Very useful for rolling up billing information



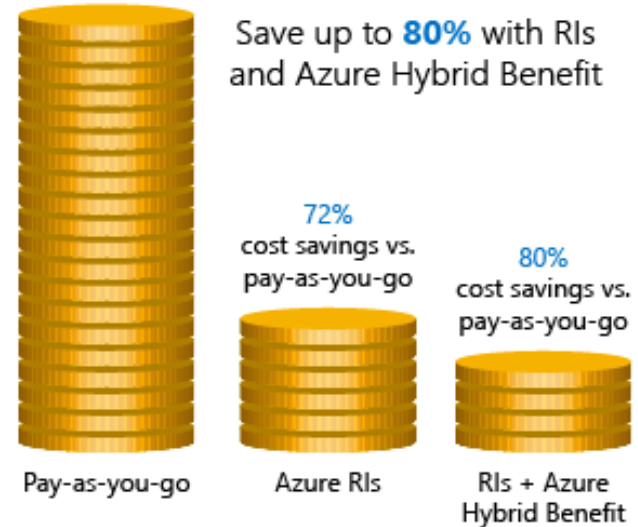
Apply Cost Savings

Azure Reservations – Helps you save money by pre-paying for services

Azure Hybrid Benefits – Use Windows Server and SQL Server on-premises licenses with Software Assurance

Azure Credits – Monthly credit benefit that allows you to experiment with, develop, and test new solutions on Azure

Regions – Choose low-cost locations and regions



Lesson 02: Configure Azure Policy





Create Management Groups



Implement Azure Policy



Create Azure Policies



Create Policy Definitions



Create Initiative Definitions



Scope the Initiative Definition



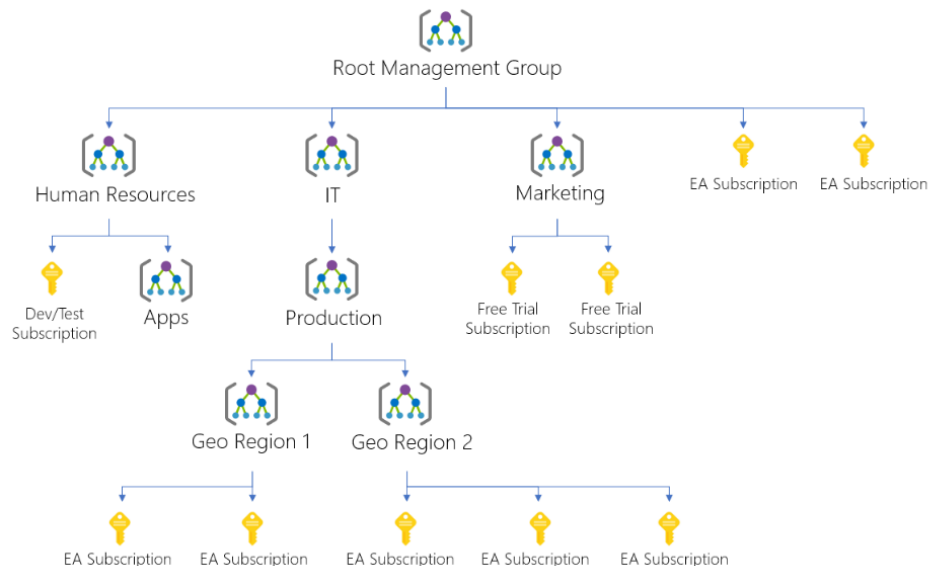
Determine Compliance

Create Management Groups

Provides a level of scope above subscriptions

Targeting of policies and spend budgets across subscriptions and inheritance down the hierarchies

Compliance and cost reporting by organization (business/teams)



Implement Azure Policies

A service to create, assign, and manage policies

Runs evaluations and scans for non-compliant resources

Advantages:

Enforcement and compliance

Apply policies at scale

Remediation

Usage Cases

Allowed resource types – Specify the resource types that your organization can deploy

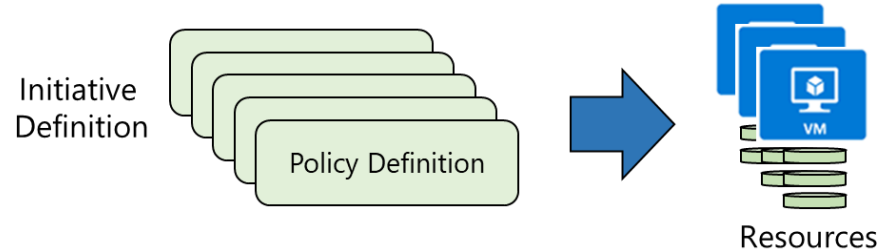
Allowed virtual machine SKUs – Specify a set of virtual machine SKUs that your organization can deploy

Allowed locations – Restrict the locations your organization can specify when deploying resources

Require tag and its value – Enforces a required tag and its value

Azure Backup should be enabled for Virtual Machines – Audit if Azure Backup service is enabled for all Virtual machines

1. Create Policy Definitions
2. Create Initiative Definitions
3. Scope the Initiative Definition
4. Determine Compliance



1. Create Policy Definitions

Many policy definitions are available

You can import policies from GitHub

Policy Definitions have a specific JSON format

You can create custom policy definitions

Policy definition

New Policy definition

BASICS

Definition location *

Visual Studio Enterprise

Name * ⓘ

Github Sample Policy

Description

A sample policy from Github.

Category ⓘ

☒ Create new ☐ Use existing

Category

POLICY RULE

↓ [Import sample policy definition from GitHub](#)

2. Create Initiative Definitions

Group policy definitions

Include one or more policies

Requires planning

Initiative definition

New Initiative definition

BASICS

Definition location *

Visual Studio Enterprise

Name * ⓘ

East Region

Description ⓘ

East Region Initiative Definition

Category ⓘ

☐ Create new ☒ Use existing

General

namingPolicyDefinition	Policy to specify allowed naming convention	Custom	Delete
regionPolicyDefinition	Policy to allow resource creation only in certain regions	Custom	Delete

3. Scope the Initiative Definition

Policy - Assignments

Search (Ctrl+/) « Assign initiative Assign policy Refresh

Scope: Visual Studio Enterprise ... Definition type: All definition types Search: Filter by name or id... Category: All categories

Total Assignments ①: 2 Initiative Assignments ①: 2 Policy Assignments ①: 0

name	↑↓ Scope	↑↓ Type	↑↓ Policies	↑↓ Category
East Region	Visual Studio Enterprise	Initiative	2	General
ASC Default (subscription: ...)	Visual Studio Enterprise	Initiative	96	Security Center

Assign the definition
to a scope

The scope enforces
the policy

Select the subscription,
and optionally the
resource group

4. Determine Compliance

 Policy - Compliance

Search (Ctrl+/)

«

Overview

Getting started

Join Preview

Compliance

Remediation

Authoring

Assignments

Definitions

Assign policy

Assign initiative

Refresh

Scope

Visual Studio Enterprise

Type

All definition types

Compliance state

All compliance states

Search

Filter by name or id...

Overall resource compliance ⓘ

98%

159 out of 162

Non-compliant initiatives ⓘ

1

out of 2

Non-compliant policies ⓘ

12

out of 98

Non-compliant resources ⓘ

3

out of 162

Name	Scope	Compliance state	Resource compli...	Non-Compliant Resources	Non-compliant policies
ASC Default (subscription: 957...	Visual Studio Enterprise	Non-compliant	98% (159 out of 162)	3	12
East Region	Visual Studio Enterprise	Not started	100% (0 out of 0)	0	0

Non-compliant initiatives

Non-compliant policies

Non-compliant resources

Lesson 03 : Configure Role-Based Access Control





Implement Role-Based Access Control



Create a Role Definition



Create a Role Assignment



Compare Azure RBAC Roles to Azure AD Roles



Apply RBAC Authentication



Determine Azure RBAC Roles

Implement Role-Based Access Control

Provides fine-grained access management
of resources in Azure

Built on Azure Resource Manager
Segregate duties within your team
Grant only the amount of access to users that
they need to perform their jobs

Concepts

Security principal. Object that represents something that is requesting access to resources

Role definition. Collection of permissions that lists the operations that can be performed

Scope. Boundary for the level of access that is requested

Assignment. Attaching a role definition to a security principal at a particular scope:

- Users can grant access described in a role definition by creating an assignment
- Deny assignments are currently read-only and are set by Azure Blueprints and Azure Managed Apps

Collection of permissions that lists the operations that can be performed

Owner

Contributor

Reader

...

Backup Operator

Security Reader

User Access Administrator

Virtual Machine Contributor

Built-in

Reader Support Tickets

Virtual Machine Operator

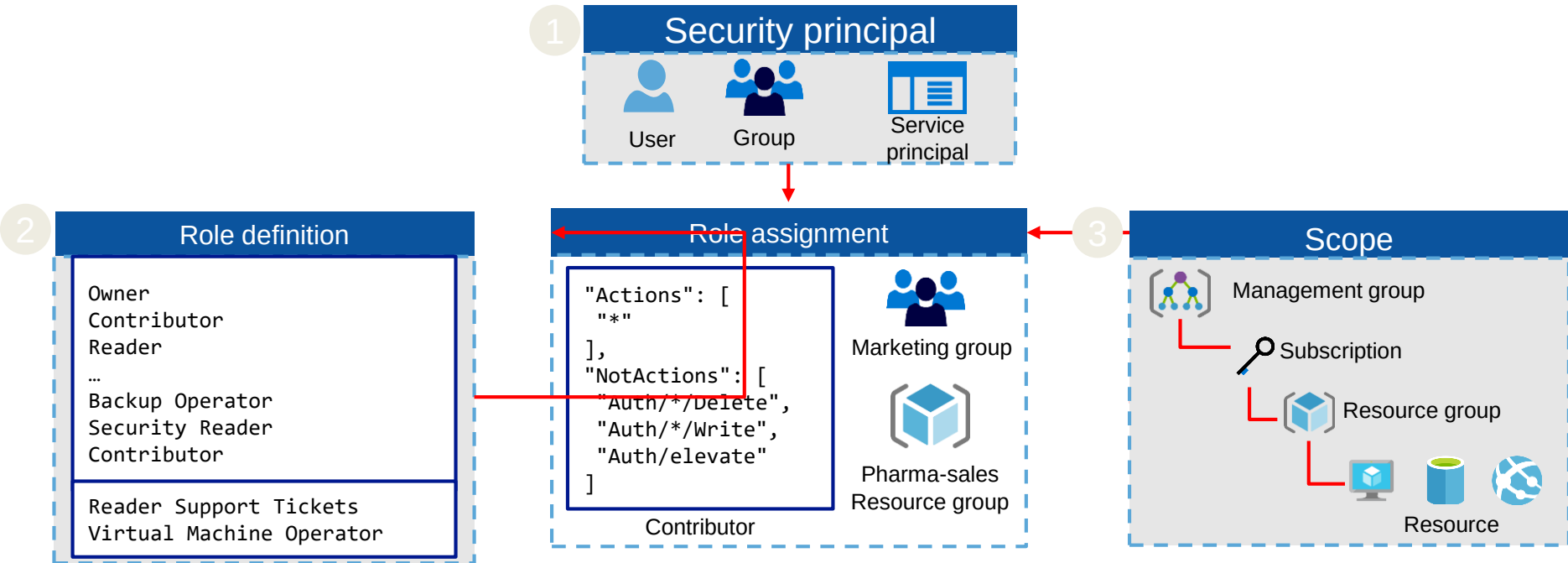
Custom

Contribut

```
"Actions": [
  "*"
],
"NotActions" : [
  "Authorization/*/Delete",
  "Authorization/*/Write",
  "Authorization/elevateAccess/Action"
],
>DataActions" : [],
"NotDataActions": [],
"AssignableScopes" : [
  "/"
]
```


Create a Role Assignment

Process of binding a role definition to a user, group, or service principal at a scope for the purpose of granting access



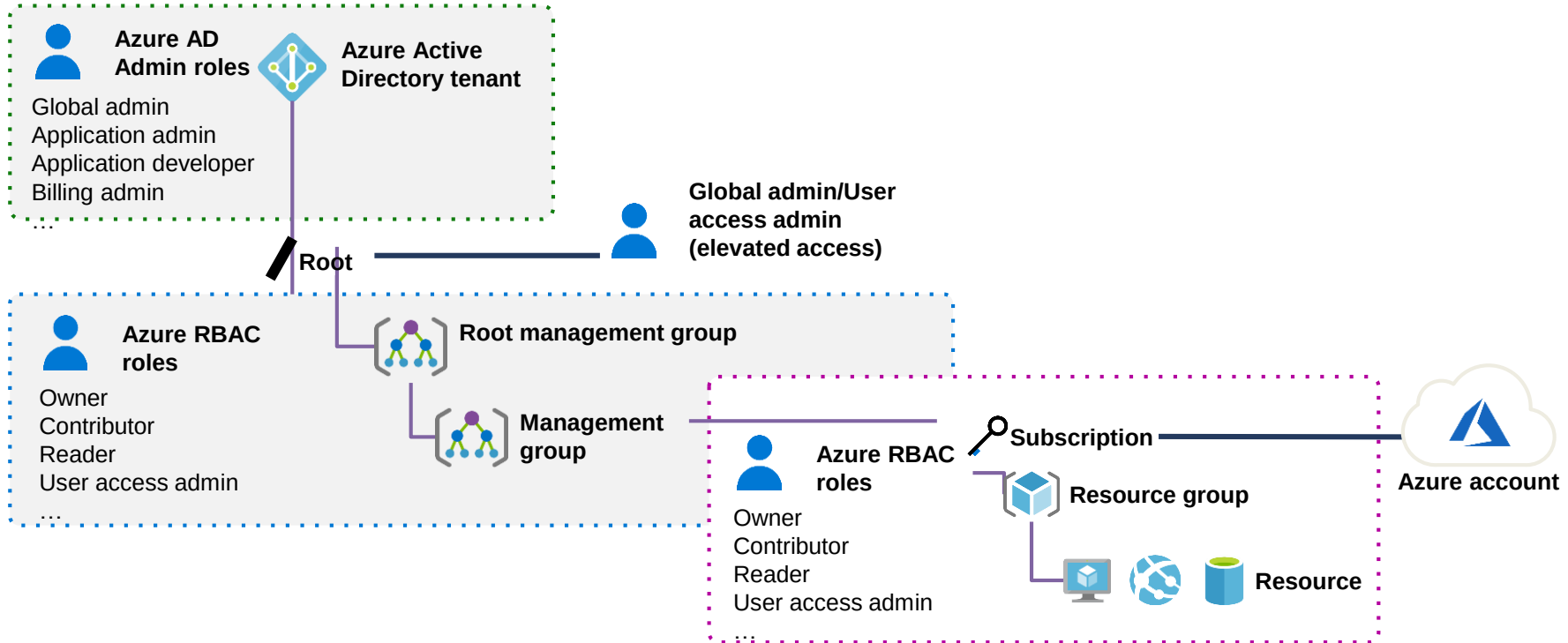
Azure RBAC Roles & Azure AD Roles

Azure RBAC roles	Azure AD roles
Manage access to Azure resources	Manage access to Azure AD objects
Scope can be specified at multiple levels	Scope is at the tenant level
Role information can be accessed in the Azure portal, Azure CLI, Azure PowerShell, Azure Resource Manager templates, REST API	Role information can be accessed in Azure portal, Microsoft 365 admin portal, Microsoft Graph, Azure Active Directory PowerShell for Graph



Classic administrator roles should be avoided if using Azure Resource Manager

Apply RBAC Authentication



Determine Azure RBAC Roles

RBAC role in Azure	Permissions	Notes
Owner	Has full access to all resources and can delegate access to others	The Service Administrator and Co-Administrators are assigned the Owner role at the subscription scope. This applies to all resource types
Contributor	Creates and manages all types of Azure resources but cannot grant access to others	This applies to all resource types
Reader	Views Azure resources	This applies to all resource types
User Access Administrator	Manages user access to Azure resources	This applies to managing access, rather than to managing resources

Thank you

